

7(a)(i)	all frequencies are amplified equally	B1
7(a)(ii)	no drop in output voltage (when there is a current)	B1
7(b)(i)	$\text{gain} = 1 + R_F / R_{IN}$ $= 1 + (4000 / 800)$	C1
	$= 6.0$	A1
7(b)(ii)	$2.0 / V_{IN} = 6.0$	A1
	$V_{IN} = (+)0.33 \text{ V}$	
7(b)(iii)	5.0 V	A1
7(b)(iv)	$V = 5.0 - 2.2 (= 2.8 \text{ V})$	C1
	$R = V / I$ $= 2.8 / 0.020$ $= 140 \Omega$	A1